

Mathematics (Type Ga)

Multiple Choice Questions

1. Let $\vec{a} = (1, -2)$ and $\vec{b} = (-1, 4)$. What is the sum of all components of the vector $\vec{a} + 2\vec{b}$? [2 points]

① 1 ② 2 ③ 3 ④ 4 ⑤ 5

2. What is the value of $\lim_{x \rightarrow 0} \frac{x^2 + 5x}{\ln(1 + 3x)}$? [2 points]

① $\frac{7}{3}$ ② 2 ③ $\frac{5}{3}$ ④ $\frac{4}{3}$ ⑤ 1

3. Consider points $A(2, a, -2)$ and $B(5, -2, 1)$ in 3-dimensional space. If the point internally dividing the line segment AB in the ratio $2 : 1$ lies on the x -axis, what is the value of a ? [2 points]

① 1 ② 2 ③ 3 ④ 4 ⑤ 5

4. Let A and B be events such that A and B^C are mutually exclusive, and

$$P(A) = \frac{1}{3} \text{ and } P(A^C \cap B) = \frac{1}{6}.$$

What is the value of $P(B)$?

(※ A^C is the complement of A .) [3 points]

① $\frac{5}{12}$ ② $\frac{1}{2}$ ③ $\frac{7}{12}$ ④ $\frac{2}{3}$ ⑤ $\frac{3}{4}$

5. Let A be the graph of the function $y = 2^x + 2$ translated m units to the right.

Let B be the graph of the function $y = \log_2 8x$ translated 2 units to the right.

Given that graphs A and B are reflections about the line $y = x$, what is the value of the constant m ?

[3 points]

- ① 1 ② 2 ③ 3 ④ 4 ⑤ 5

6. Consider the parabola $y^2 = 12x$ with focus F .

Let P be a point on the parabola such that $\overline{PF} = 9$.

What is the x -coordinate of point P ? [3 points]

- ① 6 ② $\frac{13}{2}$ ③ 7 ④ $\frac{15}{2}$ ⑤ 8

7. Consider the tangent line to the curve $e^x - xe^y = y$ at point $(0, 1)$. What is the slope of this line? [3 points]

- ① $3 - e$ ② $2 - e$ ③ $1 - e$ ④ $-e$ ⑤ $-1 - e$

8. Let X be a random variable following the binomial distribution $B\left(n, \frac{1}{2}\right)$ such that $E(X^2) = V(X) + 25$.

What is the value of n ? [3 points]

- ① 10 ② 12 ③ 14 ④ 16 ⑤ 18

9. Let $f(x) = \frac{1}{1 + e^{-x}}$, and let $g(x)$ be its inverse.

What is the value of $g'(f(-1))$? [3 points]

- ① $\frac{1}{(1+e)^2}$ ② $\frac{e}{1+e}$ ③ $\left(\frac{1+e}{e}\right)^2$
 ④ $\frac{e^2}{1+e}$ ⑤ $\frac{(1+e)^2}{e}$

10. There is a sack containing 7 marbles marked with integers from 2 to 8 respectively.

If 2 marbles were randomly taken out of this sack at the same time, what is the probability that the 2 integers marked on the marbles taken out are coprime? [3 points]

- ① $\frac{8}{21}$ ② $\frac{10}{21}$ ③ $\frac{4}{7}$ ④ $\frac{2}{3}$ ⑤ $\frac{16}{21}$

11. Suppose $0 \leq \theta < 2\pi$. The quadratic equation

$$6x^2 + (4 \cos \theta)x + \sin \theta = 0$$

has no real solutions for x , if and only if $\alpha < \theta < \beta$.

What is the value of $3\alpha + \beta$? [3 points]

- ① $\frac{5}{6}\pi$ ② π ③ $\frac{7}{6}\pi$ ④ $\frac{4}{3}\pi$ ⑤ $\frac{3}{2}\pi$

12. What is the number of ways to give 8 identical chocolates to 4 students A, B, C and D according to the following rules? [3 points]

- (a) Each student receives at least 1 chocolate.
 (b) Student A receives more chocolates than student B.

- ① 11 ② 13 ③ 15 ④ 17 ⑤ 19

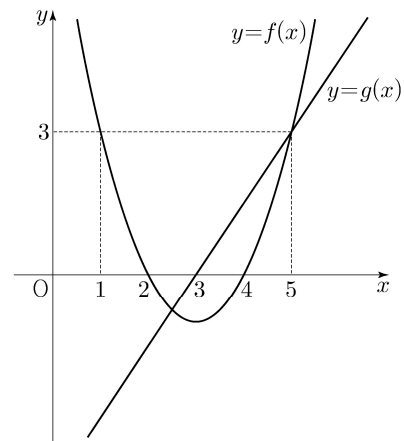
13. Consider a plane in 3-dimensional space which contains the point $(2, 0, 5)$ and the line $x - 1 = 2 - y = \frac{z+1}{2}$. What is the x -coordinate of the intersection between this plane and the x -axis?
[3 points]

- ① $\frac{9}{2}$ ② 4 ③ $\frac{7}{2}$ ④ 3 ⑤ $\frac{5}{2}$

14. The graph of a quadratic function $f(x)$ and a linear function $g(x)$ is shown below. What is the sum of all positive integers x for which

$$\left(\frac{1}{2}\right)^{f(x)g(x)} \geq \left(\frac{1}{8}\right)^{g(x)}$$

is satisfied? [4 points]



- ① 7 ② 9 ③ 11 ④ 13 ⑤ 15

15. On some day, the commuting time of employees of some company followed a normal distribution with a mean of 66.4 minutes and a standard deviation of 15 minutes.

On this day, 40% of employees with a commuting time of **73 minutes or more**, and 20% of employees with a commuting time **less than 73 minutes**, commuted by the subway.

All other employees commuted by other means.

Among the employees of this company who commuted this day, what is the probability that a randomly selected employee has commuted by the subway?

(※ For a random variable Z that follows the standard normal distribution, suppose $P(0 \leq Z \leq 0.44) = 0.17$.)

[4 points]

- ① 0.306 ② 0.296 ③ 0.286 ④ 0.276 ⑤ 0.266

16. A continuous function $f(x)$ defined on $x > 0$ satisfies

$$2f(x) + \frac{1}{x^2}f\left(\frac{1}{x}\right) = \frac{1}{x} + \frac{1}{x^2}$$

for all positive numbers x .

What is the value of $\int_{\frac{1}{2}}^2 f(x)dx$? [4 points]

- ① $\frac{\ln 2}{3} + \frac{1}{2}$ ② $\frac{2 \ln 2}{3} + \frac{1}{2}$ ③ $\frac{\ln 2}{3} + 1$
 ④ $\frac{2 \ln 2}{3} + 1$ ⑤ $\frac{2 \ln 2}{3} + \frac{3}{2}$

17. Consider the set $X = \{1, 2, 3, 4, 5, 6\}$ and a function $f : X \rightarrow X$. The following passage computes the number of functions f for which the image of the composite function $f \circ f$ has 5 elements.

Let A and B be the image of f and $f \circ f$ respectively.

If $n(A) = 6$, then f is bijective and $f \circ f$ is also bijective, so $n(B) = 6$.

If $n(A) \leq 4$, then $n(B) \leq 4$ since $B \subseteq A$.

Therefore it should be that $n(A) = 5$ and $B = A$.

(i) There are $\boxed{(\alpha)}$ ways to select a subset A of X where $n(A) = 5$.

(ii) For the set A selected in (i), let k be the element of X that is not in A . Since $n(A) = 5$, there are $\boxed{(\beta)}$ ways to select $f(k)$ from A .

(iii) Consider $A = \{a_1, a_2, a_3, a_4, a_5\}$ selected in (i) and $f(k)$ selected in (ii).

Since $f(k) \in A$ and $A = B$, it should be that $A = \{f(a_1), f(a_2), f(a_3), f(a_4), f(a_5)\}$.

The number of cases where this holds is equal to the number of bijections from A to A , which is $\boxed{(\gamma)}$.

By (i), (ii) and (iii), the number of functions f to compute is $\boxed{(\alpha)} \times \boxed{(\beta)} \times \boxed{(\gamma)}$.

Let p, q and r be the correct numbers for (α) , (β) and (γ) respectively. What is the value of $p + q + r$?

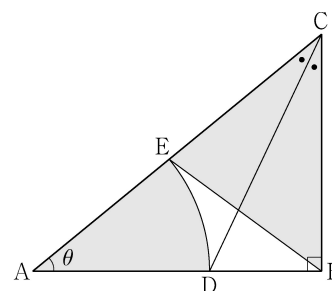
[4 points]

- ① 131 ② 136 ③ 141 ④ 146 ⑤ 151

18. As the figure shows, consider a right triangle ABC with $\overline{AB} = 1$ and $\angle B = \frac{\pi}{2}$. Let D be the intersection between line AB and a line that bisects $\angle C$. Let E be the intersection between the line segment AC and a circle with center A and radius $\overline{AD}$.

When $\angle A = \theta$, let $S(\theta)$ be the area of sector ADE , and let $T(\theta)$ be the area of triangle BCE .

What is the value of $\lim_{\theta \rightarrow 0^+} \frac{\{S(\theta)\}^2}{T(\theta)}$? [4 points]



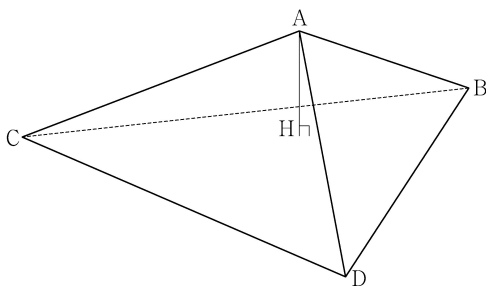
- ① $\frac{1}{4}$ ② $\frac{1}{2}$ ③ $\frac{3}{4}$ ④ 1 ⑤ $\frac{5}{4}$

19. There is a tetrahedron ABCD whose face BCD is an equilateral triangle with side lengths 12. Let H be the perpendicular foot from point A to plane BCD. It is given that point H lies inside triangle BCD, triangle CDH has 3 times the area of triangle BCH, triangle DBH has 2 times the area of triangle BCH, and $\overline{AH} = 3$.

Let M be the midpoint of the line segment BD, and let Q be the perpendicular foot from point A to line CM. What is the length of the line segment AQ?

[4 points]

- ① $\sqrt{11}$ ② $2\sqrt{3}$ ③ $\sqrt{13}$ ④ $\sqrt{14}$ ⑤ $\sqrt{15}$



20. Consider all lines passing through point $\left(-\frac{\pi}{2}, 0\right)$ that are tangent to the curve $y = \sin x$ ($x > 0$).

Let us list all the x -coordinates of the points of tangency in ascending order.

Let a_n be the n th number in this list.

Which statements in <List> are true for all positive integers n ? [4 points]

- < L i s t >
- a. $\tan a_n = a_n + \frac{\pi}{2}$
 - b. $\tan a_{n+2} - \tan a_n > 2\pi$
 - c. $a_{n+1} + a_{n+2} > a_n + a_{n+3}$

- ① a ② a, b ③ a, c
④ b, c ⑤ a, b, c

21. A function $f(x)$ differentiable on the set of all real numbers satisfies the following. What is the value of $f(-1)$? [4 points]

(A) $2\{f(x)\}^2 f'(x) = \{f(2x + 1)\}^2 f'(2x + 1)$
for all real numbers x .

(B) $f\left(-\frac{1}{8}\right) = 1$ and $f(6) = 2$.

- ① $\frac{\sqrt[3]{3}}{6}$ ② $\frac{\sqrt[3]{3}}{3}$ ③ $\frac{\sqrt[3]{3}}{2}$ ④ $\frac{2\sqrt[3]{3}}{3}$ ⑤ $\frac{5\sqrt[3]{3}}{6}$

Short Answer Questions

22. Compute ${}_6P_2 - {}_6C_2$. [3 points]

23. Given that $\tan \theta = 5$, compute $\sec^2 \theta$. [3 points]

24. Suppose a point P is moving on the xy -plane, and its position (x, y) at time t ($t \geq 0$) is equal to

$$x = 1 - \cos 4t \text{ and } y = \frac{1}{4} \sin 4t.$$

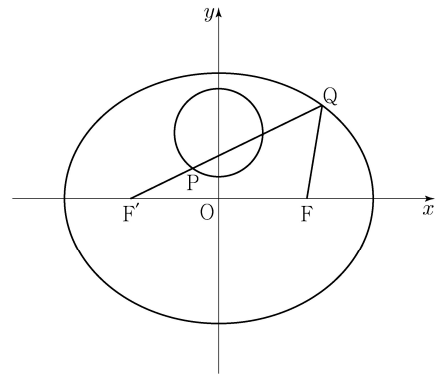
Compute the magnitude of acceleration of point P when its speed is at its greatest. [3 points]

25. Compute $\int_0^\pi x \cos(\pi - x) dx$. [3 points]

26. In some region, the daily leisure time of residents follows a normal distribution with a mean of m minutes and a standard deviation of σ minutes. Suppose 16 residents were randomly sampled from this region, and their daily leisure time had a mean of 75 minutes. The 95% confidence interval for m computed with this sample is $a \leq m \leq b$. Suppose 16 residents were randomly sampled from this region again, and their daily leisure time had a mean of 77 minutes. The 99% confidence interval for m computed with this sample is $c \leq m \leq d$. Given that $d - b = 3.86$, Compute the value of σ . (* For a random variable Z that follows the standard normal distribution, suppose $P(|Z| \leq 1.96) = 0.95$ and $P(|Z| \leq 2.58) = 0.99$.) [4 points]

27. Let us throw a die once. Let A be the event where it lands on an odd number, and let B be the event where it lands on a factor of m for some positive integer $m \leq 6$. Compute the sum of all values of m for which the events A and B are independent.
- [4 points]

28. Consider the ellipse $\frac{x^2}{49} + \frac{y^2}{33} = 1$ with two foci F and F' . For a point P on the circle $x^2 + (y - 3)^2 = 4$, let Q be the point with a positive y -coordinate where line $F'P$ meets the ellipse. Compute the greatest possible value of $\overline{PQ} + \overline{FQ}$. [4 points]



29. Consider a triangle ABC on the xy -plane with an area of 9. Let P, Q and R be points that move freely on line segments AB, BC and CA respectively.

The set of all points X that satisfy

$$\overrightarrow{AX} = \frac{1}{4}(\overrightarrow{AP} + \overrightarrow{AR}) + \frac{1}{2}\overrightarrow{AQ}$$

is a region with an area of $\frac{q}{p}$. Compute $p + q$.

(※ p and q are positive integers that are coprime.)

[4 points]

30. Let $f(x)$ be a cubic function with a leading

coefficient of 6π , and let $g(x) = \frac{1}{2 + \sin(f(x))}$.

Let us list all values of $\alpha \geq 0$ for which $g(x)$ has a local maximum or a local minimum at $x = \alpha$.

Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \dots$ be this list in ascending order. It is given that $g(x)$ satisfies the following.

(A) $\alpha_1 = 0$ and $g(\alpha_1) = \frac{2}{5}$.

(B) $\frac{1}{g(\alpha_5)} = \frac{1}{g(\alpha_2)} + \frac{1}{2}$.

Given that $g'\left(-\frac{1}{2}\right) = a\pi$, compute a^2 .

(※ $0 < f(0) < \frac{\pi}{2}$) [4 points]